ENTERPRISE AI PROPOSAL

Agent Memory Graph

Product & Deployment Proposal

Production-Ready Platform for Grounded, Auditable Multi-Agent Coordination

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A Monday Morning Without Memory

Meet Martín Díaz — and the problem AMG was built to solve.

It is 9:14 AM on a Monday. Martín Díaz, Data Engineer on the ADO squad, sits down to build a new agent. His goal is simple: an assistant that helps his team find the right dataset owner when something breaks in production. The data catalog has the answers — he just needs the agent to find them reliably.

He wires the agent to the GenAI Gateway and runs his first query: *"Who owns the flight punctuality dataset and what's the refresh schedule?"* The agent replies with a name and a schedule. Sounds right. Martín uses it to route an incident ticket.

Two hours later, the ticket bounces back. Wrong owner. The dataset was transferred to a different squad three months ago, but the agent had no way to know — it started with zero context, scraped the catalog on the fly, and returned a stale result. No memory, no continuity, no audit trail.

The cold start problem — measured:

45 minutes of Martín's time re-explaining context to the agent across three separate sessions that week

4,200 tokens spent per session on re-loading background knowledge the agent already retrieved the day before

1 misrouted incident because the agent returned a stale fact with no confidence signal or timestamp

Zero audit trail — no way to know where the wrong answer came from or which call triggered it

The Same Monday, With AMG

With Agent Memory Graph, Martín's agent starts differently. When it receives the ownership query, the MindMap pipeline fires: it mines the entity *"flight punctuality dataset"* from the prompt, traverses the Semantic graph, and retrieves 12 facts — including the current owner, the transfer event, the refresh schedule, and the last quality score — all with confidence weights and timestamps. The facts are serialized with unique IDs and injected into the synthesis prompt. The agent cites `fact-id: f_7a3c` as the source of the ownership answer.

The answer is correct. The transfer is reflected. The incident is routed in 90 seconds. The audit log shows exactly which fact the agent relied on, when it was last validated, and what its confidence weight was at query time.

With AMG — same query, different outcome:

12 facts retrieved in under 5ms — pre-weighted, pre-validated, with provenance

> **1,200 tokens** used at synthesis — only the high-confidence, relevant facts were included
>
> **Correct owner returned** — the transfer event had been ingested from DataHub and weighted appropriately
>
> **Full audit trail** — every access, every fact cited, every authorization decision logged immutably

Martín did not change how he wrote the agent. He did not add prompt engineering or manual context loading. AMG handled it at the infrastructure layer — transparently, deterministically, and with zero hallucination risk.

That is what AMG gives your organization: a persistent, shared, auditable memory layer that transforms every AI agent from a stateless call into a context-aware, evidence-grounded workflow participant.

1. Executive Summary

Agent Memory Graph (AMG) is an enterprise-grade, GCP-native AI agent coordination platform that solves the most critical failure modes of production AI deployments: context loss, hallucination, and uncontrolled inference cost. Its two core innovations — the MindMap reasoning pipeline and the Cognitive Weighting system — work together to deliver measurably better AI: lower hallucination rates, faster retrieval, and significantly reduced LLM token consumption compared to conventional RAG architectures.

The MindMap pipeline (inspired by the ACL'24 research paper by Wen, Wang & Sun) forces every agent response to be built exclusively from graph-retrieved evidence. The Cognitive Weighting system (inspired by Engram's provable memory infrastructure) assigns Bayesian confidence scores to every fact in memory, applies temporal decay to stale knowledge, and gates synthesis on freshness and confidence thresholds. The practical result: agents retrieve only high-signal, recent facts; unused or contradicted facts fade out automatically; and token context is kept bounded and relevant — avoiding the sprawling, expensive context windows typical of naive RAG.

For your organization, AMG is the persistent coordination layer for your AI infrastructure: a shared memory fabric that all agents — from internal engineering copilots to customer-facing chatbots — can trust and reason over. AMG is production-ready (v0.2, 843 tests passing, P99 < 5ms), deploys within your existing GCP organization, and integrates naturally with your GenAI platform, data catalog, and service management tools.

> **Key numbers:** P99 latency < 5ms (10x below SLO). Cognitive weighting bounds retrieval context to only high-confidence facts — directly reducing LLM token spend per agent turn. Grounding rate: 100% of responses backed by evidence or explicit insufficiency (zero hallucination).

2. Product Description

What It Does

Agent Memory Graph is a deterministic, grounding-first memory system for coordinating multiple AI agents within a single enterprise. It eliminates hallucination by forcing every agent response to be derived from evidence retrieved from a graph database. It preserves organizational context across sessions, users, and agents. And it provides immutable audit trails for every reasoning step — critical for regulated industries and compliance-sensitive workflows.

Problems It Solves

Enterprise AI deployments fail for consistent, well-understood reasons. AMG addresses each one with a specific architectural decision — not a workaround:

Problem	How AMG Fixes It	Key Feature
Context Loss — agents forget interactions across sessions; users re-explain problems repeatedly	Context tier stores full session history; Semantic tier persists organizational facts indefinitely; recall is automatic on every agent turn	Tripartite Memory Model
Hallucination — agents invent facts not grounded in organizational data	MindMap pipeline: synthesis is prohibited unless facts are retrieved and serialized first. No evidence = explicit insufficiency signal	MindMap Grounding Pipeline
High Token Cost — naive RAG floods context with loosely relevant chunks	Cognitive Weighting filters retrieval to only high-confidence, fresh facts before synthesis. Bounded traversal prevents context sprawl	Cognitive Weighting (Bayesian)
Low Retrieval Accuracy — vector similarity returns topically similar but factually unreliable chunks	Graph traversal over Semantic tier follows typed relationships between entities, producing structurally grounded paths	Spanner Graph Retrieval
Auditability Gap — no reasoning traces; impossible to debug decisions or prove compliance	Audit tier logs every memory access, tool call, authorization decision, and reasoning step immutably (append-only, never deleted)	Immutable Audit Trail
Multi-Agent Coordination Breakdown — multiple agents make conflicting decisions with no shared context	All agents share the same Semantic graph. Contradiction detection flags conflicting facts before synthesis	Shared Org Memory + Contradiction Detection

What AMG Provides

Tripartite Memory Model — Strict separation of Context (user-specific session history), Semantic (durable organizational facts with Bayesian confidence), and Audit (immutable compliance logs).

MindMap Reasoning Pipeline — A deterministic four-stage pipeline: Mine → Aggregate → Serialize → Synthesize. Every response is built exclusively from retrieved graph evidence.

Cognitive Weighting System — Bayesian log-odds weights on every node and edge, temporal decay for stale facts, confidence + freshness gating on synthesis.

Multi-Agent Coordination — All agents share the same Semantic memory graph. Coordinated, non-redundant operations with contradiction detection.

Immutable Audit Trail — Every access, tool call, and reasoning step in the Audit tier (append-only) for full compliance traceability.

ABAC Governance — Fine-grained authorization controls scope which agents can access which subgraphs and tiers. Deny-by-default with audited reason codes.

Document Extraction Pipeline — Ingest PDFs and PowerPoint files, extract entities, deduplicate, and populate the Semantic graph automatically.

3. Architecture Brief

AMG is a GCP-native Python service built on FastAPI, deployed on Cloud Run, backed by Google Cloud Spanner Graph for the memory store. All components are containerized and managed through Terraform infrastructure-as-code. The system deploys as a dedicated single-organization service within your GCP project — no data ever leaves your environment.

Memory Tiers

AMG divides all knowledge into three strictly isolated tiers with different retention policies, mutability rules, and access controls. The design mirrors human memory architecture: working memory (Context), long-term memory (Semantic), and autobiographical log (Audit).

Tier	Scope & Owner	What It Stores	Lifecycle
CONTEXT	Per user / per session	User-specific conversation turns, agent responses, session state, intermediate reasoning. Scoped strictly to one session and one user — no cross-user sharing.	24–48 hours (rapid Bayesian decay on idle sessions)
SEMANTIC	Organization-wide	Durable facts, entity relationships, discovered insights, ingested document knowledge. Shared across all agents in the organization within ABAC policy scope.	Months to years (slow decay; Bayesian reinforcement on repeated access)
AUDIT	Permanent / compliance	Every memory access, tool call, authorization grant/deny, reasoning step, and orchestration trace. Indexed with reason codes.	Permanent retention (append-only, never modified)

Privacy note: The Context tier is the right home for any user-identifiable data in agent interactions (customer IDs, employee IDs, session content). Its 24-48h decay and strict per-session scope mean personal data is not persisted indefinitely — aligned with data minimization principles.

MindMap Reasoning Pipeline

The MindMap pipeline is the core reasoning engine of AMG. It enforces a strict evidence discipline: no LLM generates answers directly from user input. Every response passes through four

deterministic stages:

Stage	What It Does	Output
1. Mine	Extracts structured intent and named entities from the agent request. Identifies query anchors for graph traversal.	Entity list + intent classification
2. Aggregate	Traverses the Spanner Graph from identified anchors. Bounded traversal depth and cognitive weight thresholds filter out stale or low-confidence paths. Only ABAC-permitted subgraphs are accessed.	Evidence path set (typed graph traversal results)
3. Serialize	Converts graph paths and edges into canonical, traceable fact payloads. Each fact receives a unique ID for citation in the final response. Provenance (source node, edge type, confidence score) is preserved.	Serialized fact payload with fact IDs
4. Synthesize	Generates the agent response ONLY from the serialized fact payload using the configured LLM. Freshness and confidence gates are checked before synthesis. If no facts exist or confidence is below threshold, returns an explicit insufficiency signal — never guesses.	Grounded response or explicit insufficiency

> **Grounding validation:** After synthesis, AMG verifies that every claim in the response references a fact ID present in the serialized evidence. Any claim referencing an unknown fact ID causes the response to fail closed — a hard guarantee against hallucination.

Cognitive Weighting System

The Cognitive Weighting system gives AMG its ability to deliver high-accuracy retrieval at low token cost. Every node and edge in the memory graph carries a LogOddsWeight that evolves through:

Mechanism	Formula / Rule	Effect
Initial weight	LogOddsWeight = 0.5 (62% confidence baseline)	New facts enter with moderate confidence — not assumed fully reliable.
Bayesian reinforcement	Weight += ln(1.25) ≈ +0.22 bits per successful access	Frequently retrieved facts gain confidence automatically. High-signal knowledge rises to the top of traversal results.
Temporal decay	Weight(t) = Weight(0) × e^(−0.05 × hours_since_access)	Unused facts fade out. A fact at weight 2.73 after heavy use decays to near-zero after one week without access.
Pruning threshold	Weight < 0.30 → soft-deleted from active traversal	Stale, low-confidence facts stop polluting retrieval results without a hard delete.
Contradiction detection	On ingest, conflicting facts are flagged with a tension marker	Synthesis is gated: contradicted facts surface as ambiguous, not authoritative.
Freshness gating	Synthesis checks average_freshness of evidence set against freshness_threshold	Responses explicitly signal when evidence is stale — preventing overconfident answers from outdated knowledge.

Business impact: Instead of sending hundreds of loosely relevant RAG chunks to the LLM, AMG sends only the high-confidence, fresh, weighted facts that passed all filters. This bounds the synthesis context window, reduces LLM token consumption per agent turn, and raises answer accuracy.

Technology Stack

Component	Technology	Notes
Runtime	Python 3.14 / FastAPI	Strict type-checking (basedpyright, 0 errors, 0 warnings)
Memory Store	Google Cloud Spanner Graph	ISO GQL queries, interleaved edge storage for fast traversal
Deployment	Cloud Run (containerized)	Terraform IaC, 3-environment strategy (dev/intg/prod)
Observability	GCP Logging + Tracing + Metrics	Structured JSON logs, OpenTelemetry traces, Cloud Monitoring
Auth & Governance	Internal ABAC or External Delegation	Pluggable: connect to Okta, Auth0, or internal policy
LLM Backend	Your GenAI Platform	Supports Claude, GPT, Gemini per your configuration
CI/CD	GitLab CI / Cloud Build	Pre-commit hooks, ruff linting, type checking, pytest — required before merge
Tests	pytest — 843 tests, 100% passing	Unit + Integration + Cross-tenant isolation + Chaos scenarios

Deployment Model

AMG is deployed as a dedicated single-organization service within your GCP organization. All agents in the organization share memory within the same company_id context. There is no multi-tenancy — data never leaks across organizations. The deployment follows a three-environment strategy managed through Terraform workspaces:

Development — Agent integration testing, API wiring, schema validation, first squad onboarding.

Integration (INTG) — Chaos engineering, load testing (7,500+ req/sec target), cross-tenant isolation validation, compliance documentation review.

Production — Canary deployment (5% traffic) → 24h monitoring → full traffic. Rollback time: < 5 minutes to previous version.

Core API Endpoints

Endpoint	Method	Purpose
`/health`	GET	Liveness and readiness probe for Cloud Run health checks
`/api/v1/agent/respond`	POST	Full multi-tool agent turn: orchestrate memory recall, graph search, evidence synthesis, and trace logging
`/api/v1/mindmap/execute`	POST	Direct MindMap pipeline execution — Mine → Aggregate → Serialize → Synthesize
`/api/v1/mcp/invoke`	POST	MCP tool invocation: memory_search, memory_add_fact, memory_recall_context, memory_log_trace
`/internal/decay-weights`	POST	Scheduled cognitive weight decay — reduces weights for idle nodes/edges
`/internal/prune-weights`	POST	Remove soft-deleted low-confidence evidence from Semantic tier

4. Strategic Value

Your organization has invested significantly in modern AI and data infrastructure. AMG is the missing coordination layer that makes these investments compound: every agent improves the shared memory; every team benefits from what others have learned.

The GenAI Ecosystem Today

From your GenAI platform, you operate a diverse model fleet (Claude, GPT-5, Gemini, Llama variants) — with per-user cost tracking and budget controls. You may also run quality feedback systems that enrich chatbot responses with human-validated knowledge. The platform demonstrates significant AI maturity.

However, these models are stateless: each call starts from scratch, with no shared context between agents, teams, or sessions. AMG directly addresses this gap — it is the persistent, shared memory layer that transforms isolated model calls into coordinated, context-aware, auditable agent workflows.

The Organizational Multiplier Effect

The compounding value of AMG grows as more agents and teams adopt it. Today each AI initiative operates in isolation — one domain's discovery goes unknown to others. AMG creates a shared organizational knowledge graph where every agent's findings enrich the corpus for all future agents.

Cognitive weighting ensures that the most frequently useful facts rise to the top automatically, while unused or contradicted knowledge decays — so the shared memory self-maintains without manual curation.

> **Compliance alignment:** If you operate in regulated environments — aviation safety, financial reporting, data protection, healthcare — AMG's immutable AUDIT tier and ABAC controls provide the traceability and access governance required for AI use in safety-critical and compliance-sensitive decisions. No simple RAG or stateless chat stack provides this.

ROI and TCO Analysis

The following figures are based on a 16-week deployment plan with 4.5 FTE, steady-state GCP infrastructure, and measurable efficiency gains from token savings and engineer time recaptured from context re-explanation. All figures are in USD.

Total Cost of Ownership — Year 1

Category	Item	Est. Cost (USD)	Notes
Engineering	4.5 FTE × 16 weeks (build + deploy)	$120,000 – $160,000	Internal engineers; no vendor licensing
Infrastructure	Cloud Run + Spanner + Build + Logging	$8,000 – $16,000/yr	~$615–$1,100/mo steady-state post go-live
GenAI Gateway	LLM inference (synthesis stage)	$6,000 – $12,000/yr	Cognitive Weighting cuts tokens ~30% vs. baseline RAG
Operations	On-call + monitoring + quarterly updates	$15,000 – $25,000/yr	0.5 FTE equivalent post go-live
Total Year 1 (build + run)		**~$149,000 – $213,000**	One-time build amortized over 3+ years
Years 2–3 (run only)		**~$29,000 – $53,000/yr**	Infrastructure + operations only

Estimated Returns — Year 1 and Beyond

Value Driver	Mechanism	Est. Annual Value (USD)
Token cost reduction	>30% fewer tokens per agent turn (Cognitive Weighting bounds context). At 500k agent turns/yr and $0.01/1k tokens average, a 30% reduction saves ~$45/day on synthesis alone.	$15,000 – $30,000/yr
Engineer time recaptured	Elimination of context re-explanation overhead. Martín's example: 45 min/week × 50 engineers × $80/hr fully-loaded = $150k/yr in recaptured engineering time.	$100,000 – $180,000/yr
Incident misrouting prevention	Each misrouted incident costs ~2–4h of cross-team resolution. At 10 incidents/month prevented, $80/hr, 3h avg: $28,800/yr avoided.	$20,000 – $40,000/yr
Compliance risk reduction	Immutable audit trail reduces AI-related compliance risk. Avoids potential fines and remediation costs from unauditable AI decisions in regulated workflows.	$50,000 – $200,000/yr (risk-adjusted)
Hallucination-driven error prevention	Zero-hallucination guarantee prevents downstream decisions based on invented facts. One critical error prevented at executive level easily exceeds the full deployment cost.	$50,000 – $500,000+ (event-driven)
Conservative total returns (Year 1)		**~$185,000 – $450,000/yr**

ROI Summary: At a Year 1 TCO of ~$160,000 (midpoint) and conservative annual returns of ~$300,000, AMG reaches **break-even at month 7–8** of operation. Over a 3-year horizon — with Years 2–3 run costs of ~$40,000/yr — the cumulative return exceeds **$700,000 against a total investment of ~$240,000**, yielding a 3-year ROI of approximately **190–400%** depending on adoption scale.

5. Use Cases Across Domains

Domain	Use Case	AMG Capability Used	Business Impact
Customer Experience	Multi-session customer chatbot with persistent context across channels	Context tier (per-customer), cross-channel Semantic facts, NPS data ingestion	Reduced handle time, higher first-contact resolution, consistent cross-channel UX
Finance	Agents querying policies and pricing without re-ingesting knowledge on every call	Semantic tier, MindMap grounding, cognitive weight prioritization	Faster analysis cycles, zero hallucinated policy rules in automated decisions
Operations	AI copilot for field technicians with safety-critical traceability for recommendations	Document extraction pipeline (technical docs), Audit tier, grounding validation	Compliance-ready AI recommendations with full reasoning audit trail
Data Governance	Data catalog-integrated assistant with persistent context about dataset ownership and lineage	Semantic tier (catalog entities), ABAC governance, Audit tier	Automated governance triage with complete auditability
Operations / Coordination	Multi-agent coordination — no agent acts on stale or contradictory information	Shared Semantic graph, contradiction detection, cross-agent context	Consistent, non-redundant operational decisions at scale
Internal Assistants	AI copilots for engineers and analysts that remember project context across weeks	Context + Semantic tier, MCP integration with Claude Code / Codeen	Faster onboarding, eliminated context re-explanation overhead

Analytics & BI	Agents querying data warehouse with cached entity relationships and transformation logic	Semantic tier (schema + lineage), cognitive weighting on query patterns	Faster queries, fewer hallucinated table/column names, lower token cost
Legal & Compliance	Compliance assistant that recalls policy precedents and traces all reasoning for audits	Semantic tier (policy documents), Immutable Audit tier with full chain-of-custody	Regulatory-ready AI decisions with complete evidence trail for auditors

6. Development & Deployment Plan

AMG v0.2 is production-ready. The deployment plan focuses on integration, adoption, and scaling — not building from scratch. The following plan is structured across four phases over approximately 16 weeks.

What We Need From Your Team

AMG is production-ready — no core R&D required. What the deployment plan needs is access, coordination time from key teams, and a focused engineering squad for integration and go-live. Below is a precise breakdown of who, how much, and what for.

Dedicated Team Composition (4.5 FTE)

Role	Allocation	Weeks Active	Primary Responsibility
Platform Engineer (GCP / Terraform)	1 FTE	1–14	Cloud Run, Spanner, IAM, Terraform workspaces, CI/CD pipeline, monitoring dashboards and SLO alerts
Backend Engineer (Python / FastAPI)	1–2 FTE	1–12	GenAI Gateway wiring, MCP registration, extraction pipeline adapters for your data catalog and knowledge base connectors
Data / AI Engineer	1 FTE	5–12	Semantic graph entity schema design, data catalog connector configuration, MAPIS ingestion pipeline setup and quality validation
Security / Governance	0.5 FTE	7–14	ABAC policy design aligned with your IAM model, compliance documentation, data residency review, audit trail sign-off
Product Manager / Lead	0.5 FTE	1–16	Pilot squad coordination, KPI definition and tracking, stakeholder communication, Phase gate reviews
QA / Integration	0.5 FTE	10–14	End-to-end chaos engineering, load testing at 7,500+ req/sec, cross-tenant isolation validation, production readiness review
Total: 4.5–5.5 FTE across 16 weeks. A lean pilot (1 use case, 1 squad) can start with 3 FTE in Phases 1–2.			

Time Required From Other Teams (Not Dedicated)

Team	Time Required	When	What For
GenAI Gateway / Cosmos Platform	~1 hour	Week 1	Issue API key for the AMG service account, confirm model access (Claude, GPT-5, Gemini), set initial monthly budget
DataHub / ATLAS	~2 hours	Week 5	Provide read token for DataHub MCP, confirm schema access for catalog connector, validate ingestion of dataset ownership and lineage facts
Network / Infrastructure	~1 business day	Week 1	Configure VPC connector for Cloud Run, open firewall rules for Spanner private IP, confirm Cloud Tasks egress routing
Security / IAM	~2 business days	Weeks 1 and 11	Review and approve service account roles, sign off on Workload Identity configuration, conduct IAM audit before PROD go-live
Legal / Compliance	~1 week	Week 11	Data residency sign-off for Spanner region, review Context tier retention policy (24–48h decay), validate ABAC design against applicable data protection requirements
Backstage / Developer Platform	~1 hour	Week 2	Register `catalog-info.yaml` in the service catalog, confirm MCP tool visibility in GenAI Hub, validate discovery from Claude Code / Codeen

GCP Access and Permissions Required

Resource	Access Required	Scope
Cloud Spanner	`roles/spanner.databaseAdmin` on the AMG project	AMG service account only
Cloud Run	`roles/run.admin` for deployments; `roles/run.invoker` for callers	AMG service account + authorized agent service accounts
Cloud Tasks	`roles/cloudtasks.enqueuer` for ingest queue	AMG service account
Cloud Build / Artifact Registry	`roles/cloudbuild.builds.editor` + `roles/artifactregistry.writer`	CI/CD service account
IAM / Service Accounts	`roles/iam.serviceAccountAdmin` to create and manage AMG service accounts	Platform engineer only, scoped to AMG project
Logging / Monitoring	`roles/logging.logWriter` + `roles/monitoring.metricWriter`	AMG service account
BigQuery (MAPIS)	`roles/bigquery.dataViewer` on `emx-data-prod-e6c5.raw_emx_mapis_rag.*`	AMG ingest service account only

APIs to enable in your GCP project: `spanner.googleapis.com` · `run.googleapis.com` · `cloudtasks.googleapis.com` · `cloudbuild.googleapis.com` · `artifactregistry.googleapis.com` · `logging.googleapis.com` · `monitoring.googleapis.com`

Required Team

Role	Qty	Responsibility
Platform Engineer (GCP)	1 FTE	Terraform, Cloud Run, Spanner, IAM, CI/CD setup, monitoring
Backend Engineer (Python)	1–2 FTE	API integration, GenAI Gateway wiring, Quality system connector, extraction pipeline adapters
Data / AI Engineer	1 FTE	Entity schema design, Semantic graph population strategy, Data catalog connectors
Security / Governance	0.5 FTE	ABAC policy design, IAM audit, compliance documentation, data residency confirmation
Product Manager / Lead	0.5 FTE	Pilot squad coordination, KPI tracking, stakeholder alignment, customer chatbot requirements
QA / Integration	0.5 FTE	End-to-end testing, chaos engineering, cross-tenant isolation validation
Total: approximately 4.5–5.5 FTE across 16 weeks. A lean pilot (1 use case, 1 team) can start with 3 FTE in Phase 1.		

Estimated Budget

Category	Item	Est. Monthly (USD)	Notes
Infrastructure	Cloud Run (2 instances, autoscale)	$150–400/mo	Scales with request volume
Infrastructure	Cloud Spanner (1 node, ~10k RU/day)	$365/mo	Scales horizontally; 1 node handles hundreds of req/sec
Infrastructure	Cloud Build / CI pipeline	$50–100/mo	Based on build frequency
Infrastructure	GCP Logging + Monitoring + Tracing	$50–200/mo	Depends on log volume and retention
LLM Inference	Your GenAI Platform (Synthesis stage)	Existing budget	Cognitive Weighting bounds context, keeping cost predictable
Engineering	Platform team (4.5 FTE × 16 weeks)	One-time ~$120–180k	Leverages internal engineers; no vendor licensing
TOTAL INFRA (monthly, steady-state post go-live): ~$615–1,100/mo *Note:* **At pilot scale (< 1k req/day), a fractional Spanner instance reduces infrastructure to ~$100–200/mo.**			

Phases & Timeline

Phase 1 — Foundation & Pilot (Weeks 1–4)

Goal: Deploy AMG in your GCP dev environment and validate the core memory pipeline with one real use case.

Week	Activity	Owner	Deliverable
1	GCP org setup: Spanner, Cloud Run, IAM, Terraform workspaces, monitoring alerts	Platform Engineer	Dev environment live
1–2	GenAI Gateway wiring to AMG Synthesize stage; smoke-test end-to-end grounded response	Backend Engineer	Full pipeline returning grounded response
2	Service catalog registration; MCP registration in your GenAI Hub	Platform Engineer	AMG visible in Backstage + discoverable from Claude Code / Codeen
2–3	First pilot squad onboarding with one use case; 50+ agent interactions in dev	Product Lead + Backend	Pilot validated in dev
4	Phase 1 review: grounding quality, P99 latency, token cost vs. baseline	All	Go/No-Go sign-off for Phase 2

Phase 2 — Integration & Data Enrichment (Weeks 5–9)

Goal: Enrich the Semantic graph with organizational data, integrate your data catalog and quality systems, expand to 2–3 pilot teams.

Week	Activity	Owner	Deliverable
5–6	Data Catalog connector: ingest dataset metadata, ownership, lineage into Semantic tier	Data/AI Engineer	Catalog entities queryable from AMG
6–7	Document extraction pipeline for knowledge bases and internal documentation	Backend + Data Engineer	First knowledge corpus ingested and weighted
7	Quality Feedback System connector: validated Q&A; flows into Semantic tier	Backend Engineer	Feedback knowledge available to all agents
7–8	ABAC policy design aligned with your IAM / identity model	Security / Governance	Authorization policy document signed off
8–9	Expand to 2–3 teams: internal tools pilot + one customer-facing use case	Product Lead	Multi-team pilot live
9	Phase 2 review: data coverage, auth compliance, coordination between teams	All	Sign-off for Phase 3

Phase 3 — Hardening & Production (Weeks 10–14)

Goal: Deploy to INTG and PROD, complete compliance documentation, certify for regulated use cases.

Week	Activity	Owner	Deliverable
10–11	INTG deployment: chaos engineering, load testing (7,500+ req/sec), cross-tenant isolation validation	Platform + QA	INTG environment validated
11–12	Compliance documentation: audit trail review, ABAC policy audit, data residency confirmation	Security / Governance	Compliance sign-off document
12–13	PROD canary deployment (5% traffic) → 24h monitoring → full traffic	Platform Engineer	PROD environment live
13–14	Runbook, escalation procedures, on-call setup, SLO dashboard in Cloud Monitoring	Platform + Product	Operational runbook published
14	Phase 3 review: availability, audit completeness, chatbot quality	All	Production sign-off

Phase 4 — Scale & Expansion (Weeks 15–16+)

Onboard all interested teams with self-service documentation and pre-seeded domain knowledge graphs.

Expand customer-facing chatbot to additional channels using the shared Context + Semantic tier for cross-channel continuity.

Evaluate publishing AMG as a reusable platform component available to all teams via your service catalog.

Define v0.3 roadmap: embedding-based retrieval, vector search over the Semantic tier, agentic workflow templates, and NPS-driven personalization.

Risk Register

Risk	Likelihood	Impact	Mitigation
Cloud Spanner quota delays	Low	Medium	Request quota increase early in Phase 1; Spanner emulator available for local dev immediately
Customer data privacy requirements (GDPR/PDPA, etc.)	Medium	High	Context tier has 24-48h decay; no persistent PII in Semantic tier by policy; ABAC controls access; legal review in Phase 3
ABAC policy complexity growth	Medium	Low	Start deny-by-default for all agents, then grant access incrementally. Keep policies declarative.
Semantic graph data quality (garbage in, garbage out)	Medium	High	Use data catalog quality scores to gate ingestion; extraction pipeline includes fuzzy dedup and confidence scoring
LLM cost overrun at chatbot scale	Low	Medium	Cognitive weighting bounds context per request; Your GenAI Platform provides per-user budget tracking and hard limits
Team availability (lean team)	High	Medium	Phase 1 can run with 3 FTE; AMG is already built — no core R&D needed. Scope pilot tightly before scaling team

7. Success Metrics

Metric	Target	Measurement
Grounding Rate	100% of responses backed by evidence or explicit insufficiency	Audit tier: grounded flag per response
Hallucination Rate	0% (no unsupported claims pass the grounding validation gate)	Automated grounding check in synthesis stage + spot audit review
Token Cost Reduction	>30% reduction vs. baseline RAG per agent turn	Your GenAI Platform per-turn token logs before vs. after AMG adoption
P99 Latency	< 50ms end-to-end (current measured baseline: < 5ms)	Cloud Monitoring percentile metrics
Availability	99.9% uptime	Cloud Run health checks + uptime alert in Cloud Monitoring
Audit Coverage	100% of authorization decisions logged with reason code	Audit tier query: granted/denied events per day
Team Adoption	5+ teams using AMG by end of Phase 4	MCP invocation count per team via GenAI Gateway logs
Customer Experience Improvement	NPS improvement ≥ 5 points vs. stateless chatbot baseline	Your data systems (NPS tracking)
Cost per Request	< $0.01 per agent turn (infra + LLM)	Your GenAI Platform cost logs + Spanner usage metrics combined

8. Recommendation & Next Steps

Agent Memory Graph is production-ready today. The recommended approach is to begin with a focused 4-week pilot led by your GenAI team on a single high-value, well-scoped use case — either an internal AI assistant or an operation-support tool. Both have clear data sources already cataloged, clear quality metrics, and limited customer data privacy implications, making them ideal for a first validation before expanding to external-facing chatbots.

Pilot Week-by-Week:

Week 1: Platform engineer deploys GCP dev environment; GenAI Gateway wired to AMG.

Week 2: GenAI team runs first agent interactions through the MindMap pipeline; validate grounding and audit trail completeness.

Week 3: Security/governance stakeholder reviews ABAC configuration and audit logs.

Week 4: Go/No-Go decision. If pass thresholds → proceed to full 16-week plan. If not → identify root cause and iterate before expanding.

The customer-facing use case — the highest-scale and highest-impact application — should enter the plan in Phase 2, after the internal pilot has validated the core pipeline and governance model. This sequencing manages risk while preserving the timeline for customer-facing value delivery.
Next Step: Schedule an initial architecture review with your platform, data, and security teams to align on deployment topology, security posture, ops model, and pilot scope.

Agent Memory Graph — Product & Deployment Proposal
Production-Ready Platform for Grounded, Auditable Multi-Agent Coordination

This proposal is provided as-is for evaluation and planning purposes. For deployment, security review, and compliance questions, please contact your solution architect.